

1(a)	units of F : kg m s^{-2}	C1
	units of r : m and units of v : m s^{-1}	A1
	units of η : $\text{kg m s}^{-2} / (\text{m} \times \text{m s}^{-1}) = \text{kg m}^{-1} \text{s}^{-1}$	
1(b)	viscosity = $0.096 / (6 \times \pi \times 0.03 \times 2.0)$	C1
	= $0.085 \text{ kg m}^{-1} \text{s}^{-1}$	A1
1(c)	one arrow vertically downwards labelled weight / W	B1
	arrow(s) vertically upwards labelled U / upthrust and drag / F_D / viscous force	B1
1(d)(i)	$V = (4 / 3) \pi r^3$	C1
	upthrust = $(4 / 3) \times \pi \times 0.03^3 \times 920 \times 9.81 = 1.0 \text{ N}$	A1
1(d)(ii)	weight = $1.0 + 0.096 (= 1.096 \text{ N})$	C1
	$m = 1.096 / 9.81$ = 0.11 kg	A1